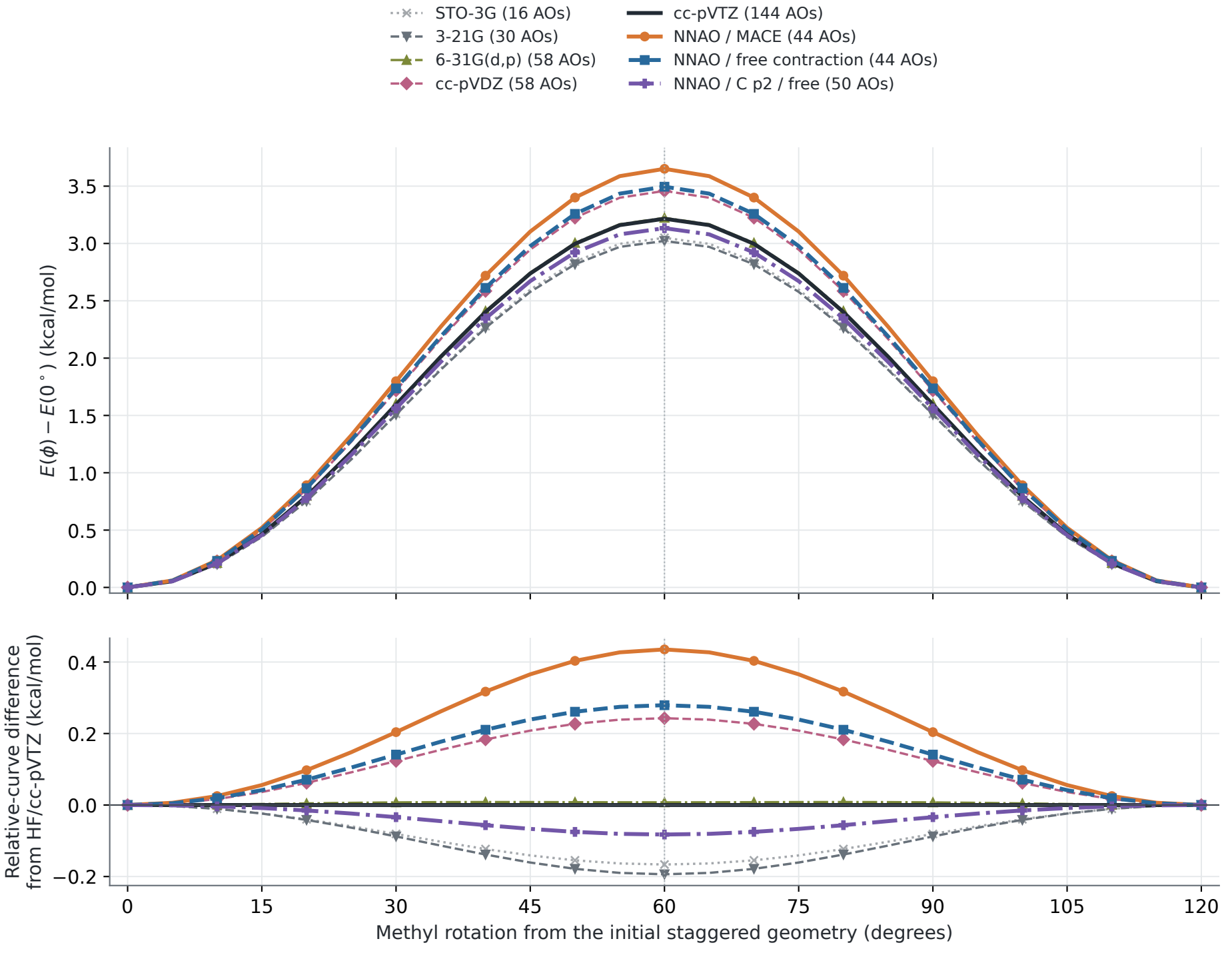
Ethane torsion: adding a second carbon valence-p radial channel
STO-3G (16 AOs)
3-21G (30 AOs)
6-31G(d,p) (58 AOs)
cc-pVDZ (58 AOs)
cc-pVTZ (144 AOs)
NNAO / MACE (44 AOs)
NNAO / free contraction (44 AOs)
NNAO / C p2 / free (50 AOs)
$E(\phi) - E(0^\circ)$ (kcal/mol)
Relative-curve difference from HF/cc-pVTZ (kcal/mol)
Methyl rotation from the initial staggered geometry (degrees)
25 rigid geometries; DF-RHF / def2-universal-jkfit. Each relative curve uses its own 0-degree energy.
New curve: C [2s2p1d], H [1s1p], free contractions; unchanged exponents, core and primitive pool. MACE remains the 44-AO checkpoint.